

TrES-1b

R-1059

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_151009 · created 2026-07-30T11:35:10+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457304.6697 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.103 +/- 0.012
Transit depth (Rp/Rs)^2	1.06 +/- 0.25 [%]
Orbital Inclination (inc)	89.28 +/- 0.95
Airmass coefficient 1 (a1)	1.169 +/- 0.01
Airmass coefficient 2 (a2)	-0.108 +/- 0.0041
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	None
Optimal Aperture	3.99
Optimal Annulus	10.84
Transit Duration (day)	0.0994 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.103 $\pm$ 0.012 vs 0.1358 (deviation 2.49 σ)
Duration fit vs published	0.0994 d vs 0.1042 d (deviation 1.61 σ)
Mid-time O-C	-3.7 min
Depth SNR	7.8
Residual scatter	0.87 %

Light curve

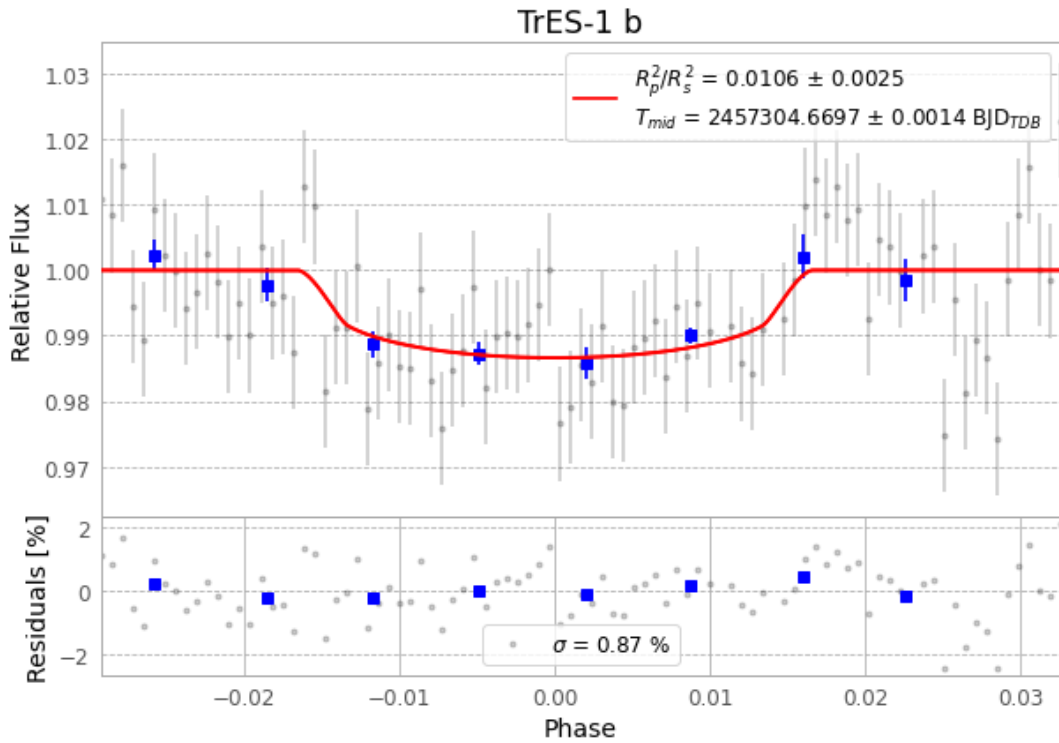


Figure 1 — FinalLightCurve_TrES-1 b_2015-10-08.png (SHA-256 5495b29d7c54fd95...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [232, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 291e70a7370b0797...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), TRES-1151009015414.FITS → TRES-1151009062412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-151009.log (c77f6d20dd06...), FinalLightCurve_TrES-1 b_2015-10-08.png (5495b29d7c54...), FinalLightCurve_TrES-1 b_2015-10-08.csv (ccca944a1bb4...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 11:35Z.